

Capstone Project Phase 1 • Fall Semester 2026

True Homeownership & Cost-of-Living Predictor

A Hyper-Local Real Estate Affordability Platform

Team Members

Brandon Matias

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Course & Faculty

Professor: Sitaram Ayyagari

Course: ITAI-2277 AI Resources &
References (CRN 21496)

Project Selection Process

Ideas on the Table

- ▶ **Smart Agriculture Sorter:** CV and robotics system for classifying produce ripeness/defects.
- ▶ **Gaming Highlight Extractor:** Automated post-match video analyzer detecting top gameplay reels.
- ▶ **Cost-of-Living Predictor:** Data-driven platform forecasting total itemized expenses based on ZIP code to determine true affordability. (Selected)

Finalization Criteria

- ▶ **16-Week Feasibility:** Preference for projects avoiding early data collection or hardware delays.
- ▶ **Demo Impact:** Must yield a high-quality, practical presentation (e.g., interactive web app).
- ▶ **Technical Diversity:** Integration of Data Engineering, Tabular ML, and Web UI disciplines.
- ▶ **Hardware Independence:** Low to zero physical hardware dependencies to minimize project risk.

Project Idea & Use Case

The Concept: Home buyers understand mortgage rates, but often overlook day-to-day carrying costs. Our platform predicts the cost of living for select ZIP codes.

The system aggregates location-based data to estimate utility expenses (water, electric), insurance requirements (flood, storm), and regional grocery indexes.

Scope & Implementation

- ▶ **Industry / Sector:** Real Estate, PropTech & Consumer Finance.
- ▶ **Geographic Scope:** Nationwide (United States), localized at the ZIP code level.
- ▶ **Target Audience:** Prospective homeowners verifying total financial affordability.



Technical Architecture & Stack



Data Sources

Aggregated public tax records, municipal utility rate datasets, FEMA flood maps, and regional Consumer Price Index (CPI) data.



Model Category

Multi-Output Tabular Regression for financial forecasting and Geospatial Risk Analysis mapping environmental insurance needs.



User Interface

Interactive web dashboard developed using Python frameworks (**Flask** or **Streamlit**) for user-friendly, real-time geographic queries.



Hardware / Infra

Standard development laptops (no physical hardware required). Pipeline and modeling facilitated by free cloud compute (**Google Colab / GCP**).

Project Plan & Resource Allocation

Team Member	Role	Primary Tasks	Resources
Khadijah Patience	Data Engineer	Sourcing tax, utility, and insurance datasets; building data pipelines.	Python Pandas Public APIs
Jonah Joseph	ML & Data Specialist	Developing Tabular Regression models for cost prediction & risk logic.	Scikit-Learn XGBoost
Brandon Matias	Full-Stack UI Dev	Building the web dashboard, integrating models via API, ensuring UX.	Flask/Streamlit Local PC
Taki Eddine Boubekri	Project Manager & QA	System integration, version control, end-to-end testing, and docs.	GitHub Jira/Trello

Questions?

Thank you for your time.